

Snehasish Nandy

Curriculum Vitae

☎ +91-9609850979
✉ snehasish@phy.nits.ac.in

PERSONAL INFORMATION

Name : Snehasish Nandy
Date of Birth : September 08, 1991
Nationality : Indian
Marital Status : Married
Address : Batrish Bigha, Jamalpur, Purba Bardhaman, West Bengal-713408, India
Email : snehasishphys@gmail.com, snehasish@phy.nits.ac.in

RESEARCH INTERESTS

- Quantum Geometry Driven Transport Phenomena in Topological Systems.
- Topological Superconductivity.
- Non-Hermitian Systems.

CURRENT POSITION

December 2023– Present **Assistant Professor**
Department of Physics
National Institute of Technology Silchar

RESEARCH POSITION

March 2022 – December 2023 **Director's Postdoctoral Research Fellow**
Physics of Condensed Matter and Complex System's Group (T-4)
Theoretical Division, Los Alamos National Laboratory, Los Alamos, USA
Mentor: Jian-Xin Zhu (email id:jxzhul@lanl.gov)

July 2019 – March 2022 **Postdoctoral Research Associate**
Department of Physics
University of Virginia, Charlottesville, USA
Mentor: Dmytro Pesin (email id:dp7bx@virginia.edu)

EDUCATION

June 2014 – July 2019 **Doctor of Philosophy in Physics** (Joint M.Sc-PhD Programme)
Department of Physics
Indian Institute of Technology Kharagpur, Kharagpur, West Bengal 721302, India
Thesis Title: Theoretical Investigations on Anomalous Transport in Topological Systems
Supervisor: Arghya Taraphder (email id:arghya@phy.iitkgp.ac.in)
Degree Obtained: August 27, 2019

June 2012 – June 2014 **Master of Science in Physics** (Joint M.Sc-PhD Programme)
Department of Physics
Indian Institute of Technology Kharagpur, Kharagpur, West Bengal 721302, India
Project Title: Catenoidal Soap Film in Electric Field
Supervisor: Sayan Kar (email id:sayan@phy.iitkgp.ac.in)
Degree Obtained: August 27, 2019 (GPA=8.76)

May 2009 – June 2012 **Bachelor of Science in Physics**
Department of Physics, Serampore College,
University of Calcutta, Kolkata, West Bengal 700016, India.
Degree Obtained: June 05, 2012 (First Class (71.50))

ACADEMIC VISIT

February 2018 – **Guest Scientist**

July 2018 Max Planck Institute for the Physics of Complex Systems
Nöthnitzer Str. 38, 01187 Dresden, Germany

Project Title: Disorder Effect in Nonlinear Hall Effect Using Quantum Boltzmann Framework

Advisor: Inti Sodemann (email id:sodemann@pks.mpg.de)

COMPUTATIONAL SKILLS

Languages: Fortran, Python, Bash scripting

Operating systems: Linux (Basic administration): Ubuntu

Scientific Packages: MATLAB, MATHEMATICA, WIEN2K, VASP, KWANT.

EDUCATIONAL ACHIEVEMENTS

2015: Qualified in National Eligibility Test (NET) in Physical Science (**all India rank 32**).

2014: Qualified in Graduate Aptitude Test in Engineering (GATE) in Physical Science (**all India rank 101**).

2012: Qualified in JAM (Joint Admission test for Masters) in Physical Science (**all India rank 215**).

FELLOWSHIP/SCHOLARSHIPS

2022-2023 Director's Fellowship at Los Alamos National Laboratory.

2022-2023 CNLS's Fellowship at Los Alamos National Laboratory.

2014-2019 Institute Research Fellowship, MHRD, Govt. of India.

2012-2014 Merit Cum Means Scholarship, Indian Institute of Technology Kharagpur.

PROJECTS

July 2025 -Present **Prime Minister Early Career Research Grant**

Title: Exploring novel transport induced by quantum geometry in unconventional superconductors

Budget: 39 Lakhs

Duration: 3 Years

Role: PI

July 2025 **Indo-Russia Joint Research Grant (Approved)**

Title: Harnessing 2D van der Waals Semiconducting Heterostructures for Quantum Optoelectronic Devices

Budget: 1 Crore

Duration: 2 Years

Role: Co-PI

GRADUATE STUDENTS MENTORED

Ritwik Roy (July 2024 - Present)

Institute Fellowship

Department of Physics

National Institute of Technology Silchar

GRADUATE STUDENTS CO-MENTORED

1. **Suvendu Ghosh** (Indian Institute of Technology Kharagpur)
Co-mentored for 3 projects. (3 Publications)
2. **Abhirup Roy Karmakar** (Center for Theoretical Studies, IIT Kharagpur)
Co-mentored for 2 projects. (2 Publications)
3. **Chuanchang Zeng** (Clemson University)
Co-mentored for 4 projects. (5 Publications)
4. **Srimayi Korrapati** (Clemson University)
Co-mentored for 2 project. (2 Publications)

5. **Supriyo Ghosh** (University of Virginia)
Co-mentored for 1 project. (1 Publication)
6. **Neelanjan Chakrabarty** (Indian Institute of Technology Kanpur)
Co-mentored for 4 projects. (1 arXiv, 3 in Preparation)

TEACHING EXPERIENCE

TEACHER

1. 'B. Tech. Engineering Physics' at NIT Silchar. (3 Semesters)
2. 'Physics of Semiconductor' course at NIT Silchar. (2 Semesters)
3. 'Mathematical Methods-I' course at NIT Silchar. (1 Semester)
4. B. Tech. Physics Laboratory at NIT Silchar. (3 Semesters)
5. M. Sc. Physics Laboratory at NIT Silchar. (3 Semester)

TEACHING ASSISTANT

1. B. Tech. Engineering Physics at IIT Kharagpur. (1 Semester)
2. 'Magnetism and Superconductivity' course at IIT Kharagpur. (1 Semester)
3. 'Advanced Condensed Matter Physics' course at IIT Kharagpur. (1 Semester)
4. B. Tech. Physics First year Laboratory at IIT Kharagpur. (2 Semesters)
5. M. Sc. Nuclear and Particle Physics Laboratory at IIT Kharagpur. (2 Semesters)
6. M. Sc. Electromagnetism and Optics Laboratory at IIT Kharagpur. (1 Semester)
7. M. Sc. Condensed Matter Physics Laboratory at IIT Kharagpur. (1 Semester)
8. M. Sc. Electronics Laboratory at IIT Kharagpur. (1 Semester)

PUBLICATIONS (Total Citation: 1484, h-index: 20, i10-index: 28)

Peer-Reviewed

1. "Gauge field induced skin effect in spinful non-Hermitian systems"
Moirangthem Sanahal, Subhasis Panda, and **S. Nandy**, Phys. Rev. B **112**, 125149 (2025).
2. "Role of disorder in the third-order anomalous Hall effect in time-reversal symmetric systems"
Chanchal K Barman, Arghya Chattopadhyay, Surajit Sarkar, Jian-Xin Zhu, and **S. Nandy**, Phys. Rev. B **111**, 235113 (2025). (Citation: 1)
3. "Fingerprint of non-Hermiticity in a d-wave altermagnet"
Gaurab Kumar Dash, Subhasis Panda, and **S. Nandy**, Phys. Rev. B **111**, 155119 (2025). (Citation: 9)
4. "Topological Optical Pseudospin Injection Beyond Weyl semimetals"
Suvendu Ghosh, Chuanchang Zeng, A Taraphder, Jian-Xin Zhu, and **S. Nandy**, Phys. Rev. B **111**, L041402 (2025).
5. "Nonlinear anomalous transverse responses induced by the Berry curvature quadrupole in systems with broken time-reversal symmetry"
Srimayi Korrapati, **S. Nandy**, and Sumanta Tewari, Phys. Rev. B **110**, 195119 (2024). (Citation: 4)
6. "Layer-dependent electronic structures and magnetic ground states of polar-polar $\text{LaVO}_3/\text{KTaO}_3$ (001) interfaces"
Shubham Patel, Narayan Mohanta, **S. Nandy**, Subhendra D. Mahanti, A Taraphder, Phys. Rev. B **110**, 054402 (2024).
7. "Quantum Hall Effect in a Weyl-Hubbard Model: Interplay Between Topology and Correlation"
S. Nandy, Christopher Lane, and Jian-Xin Zhu, Phys. Rev. B **109**, 085111 (2024). (Citation: 6)
8. "Signature of nodal topology in nonlinear quantum transport across junctions in Weyl and multi-Weyl semimetals"
Suvendu Ghosh, **S. Nandy**, Jian-Xin Zhu and A. Taraphder, Phys. Rev. B **109**, 045437 (2024). (Citation: 3)

9. “*Theoretical investigations on Kerr and Faraday rotations in topological multi-Weyl Semimetals*”
Supriyo Ghosh, Ambaresh Sahoo, and **Snehasish Nandy**, SciPost Physics **15**, 133 (2023).
(Citation: 14)
10. “*Third-order Hall effect in the surface-states of topological insulator*”
Tanay Nag, Sanjib Kumar Das, Chuanchang Zeng, and **Snehasish Nandy**, Phys. Rev. B **107**, 245141 (2023). (Citation: 24)
11. “*Decoupling intranode and internode scattering in Weyl fermions*”
Girish Sharma, **S. Nandy**, Karthik V. Raman and Sumanta Tewari, Phys. Rev. B **107**, 115161 (2023). (Citation: 20)
12. “*Quantum oscillations of the nonlinear planar effects signifying chiral anomaly in Weyl Semimetals*”
Chuanchang Zeng, **Snehasish Nandy**, Pu Liu, Sumanta Tewari, and Yugui Yaon, Phys. Rev. B **107**, L081107 (2023). (Citation: 10)
13. “*Giant anomalous thermal Hall effect in tilted type-I magnetic Weyl semimetal $\text{Co}_3\text{Sn}_2\text{S}_2$* ”
Abhirup Roy Karmakar, **Snehasish Nandy**, A. Taraphder, and G. P. Das, Phys. Rev. B **106**, 245133 (2022). (Citation: 16)
14. “*Nonreciprocal optics and magnetotransport in Weyl metals as signatures of band topology*”
S. Nandy and D.A. Pesin, Phys. Rev. B **106**, L041108 (2022). (Citation: 13)
15. “*Higher-order topological phases on fractal lattices*”
Sourav Manna, **S. Nandy**, and Bitan Roy, Phys. Rev. B **105**, L201301 (2022). (Citation: 71)
16. “*Low-energy effective theory and anomalous Hall effect in monolayer WTe_2* ”
S. Nandy, and D. A. Pesin, SciPost Phys. **12**, 120 (2022). (Citation: 9)
17. “*Chiral anomaly induced nonlinear Nernst and thermal Hall effects in Weyl semimetals*” (**Editor’s Suggestion**)
Chuanchang Zeng*, **S. Nandy***, and Sumanta Tewari, Phys. Rev. B **105**, 125131 (2022). (* equal contribution) (Citation: 28)
18. “*Chiral anomaly induced nonlinear Hall effect in semimetals with multiple Weyl points*”
S. Nandy, Chuanchang Zeng, and Sumanta Tewari, Phys. Rev. B **104**, 205124 (2021). (Citation: 39)
19. “*Topological Magnus responses in two- and three-dimensional systems*”
Sanjib Kumar Das, Tanay Nag, and **S. Nandy**, Phys. Rev. B **104**, 115420 (2021). (Citation: 21)
20. “*Nonlinear transport in Weyl semimetals induced by Berry curvature dipole*”
Chuanchang Zeng, **S. Nandy**, and Sumanta Tewari, Phys. Rev. B **103**, 245119 (2021). (Citation: 82)
21. “*Nonlinear transport without spin-orbit coupling or warping in two-dimensional Dirac semimetals*”
Sai Satyam Samal, **S. Nandy**, and Kush Saha, Phys. Rev. B **103**, L201202 (2021). (Citation: 27)
22. “*Probing mirror anomaly and classes of Dirac semimetals with circular dichroism*”
Abhirup Roy Karmakar, **S. Nandy**, G. P. Das, and Kush Saha, Phys. Rev. Research **3**, 013230 (2021). (Citation: 7)
23. “*Chiral Magnetic Effect of Hot Electrons*”
S. Nandy, and D. A. Pesin, Phys. Rev. Lett. **125**, 266601 (2020). (Citation: 16)
24. “*Magneto-transport phenomena of type-I multi-Weyl semimetals in co-planar setups*”
Tanay Nag and **S. Nandy**, Journal of Physics: Condensed matter **33**, 075504 (2020). (Citation: 50)
25. “*Sign of longitudinal magnetoconductivity and the planar Hall effect in Weyl semimetals*”
Girish Sharma, **S. Nandy**, and Sumanta Tewari, Phys. Rev. B **102**, 205107 (2020). (Citation: 36)
26. “*Fundamental relations for anomalous thermoelectric transport coefficients in the nonlinear regime*”
Chuanchang Zeng, **S. Nandy**, and Sumanta Tewari, Phys. Rev. Research **2**, 032066(R) (2020). (Citation: 50)

27. “Chirality-dependent planar Hall effect in inhomogeneous Weyl semimetals”
Suvendu Ghosh, Debabrata Sinha, **S. Nandy**, and Arghya Taraphder, Phys. Rev. B **102**, 121105(R) (2020). (Citation: 45)
28. “Dynamic chiral magnetic effect and anisotropic natural optical activity of tilted Weyl semimetals”
Urmimala Dey, **S. Nandy**, and Arghya Taraphder, Scientific Reports **10**, 2699 (2020). (Citation: 23)
29. “Generalized triple-component fermions: Lattice model, Fermi arcs, and anomalous transport”
S. Nandy, Sourav Manna, Dumitru Calugaru, and Bitan Roy, Phys. Rev. B **100**, 235201 (2019). (Citation: 33)
30. “Nonlinear Nernst effect in bilayer WTe₂”
Chuanchang Zeng, **S. Nandy**, and Sumanta Tewari, Phys. Rev. B **100**, 245102 (2019). (Citation: 62)
31. “Symmetry and quantum kinetics of the nonlinear Hall effect” (**Editor’s Suggestion**)
S. Nandy, and Inti Sodemann, Phys. Rev. B **100**, 195117 (2019). (Citation: 170)
32. “Planar thermal Hall effect in Weyl semimetals”
S. Nandy, Arghya Taraphder, and Sumanta Tewari, Phys. Rev. B **100**, 115139 (2019). (Citation: 34)
33. “Mirror anomaly and anomalous Hall effect in type-I Dirac semimetals”
S. Nandy, Kush Saha, Arghya Taraphder, and Sumanta Tewari, Phys. Rev. B **99**, 075116 (2019). (Citation: 11)
34. “Ultrafast time-resolved investigations of excitons and biexcitons at room temperature in layered WS₂”
Rup K. Chowdhury, **S. Nandy**, et. al., 2D Mater. **6**, 015011 (2018). (Citation: 31)
35. “Berry phase theory of planar Hall effect in topological insulators”
S. Nandy, Arghya Taraphder, and Sumanta Tewari, Scientific Reports **8**, 3995 (2018). (Citation: 85)
36. “Suppression of the Hall number due to charge density wave order in high- T_c cuprates”
Girish Sharma, **S. Nandy**, Arghya Taraphder, and Sumanta Tewari, Phys. Rev. B **97**, 195153 (2018). (Citation: 9)
37. “Nature of spiral state and absence of electric polarisation in Sr-doped YBaCuFeO₅ revealed by first-principle study”
Dibyendu Dey, **S. Nandy**, Tulika Maitra, C. S. Yadav, and Arghya Taraphder, Scientific Reports **8**, 1536 (2018). (Citation: 26)
38. “Chiral Anomaly as the Origin of the Planar Hall Effect in Weyl Semimetals”
S. Nandy, Girish Sharma, Arghya Taraphder, and Sumanta Tewari, Phys. Rev. Lett **119**, 176804 (2017). (Citation: 375)
39. “Anomalous transport near the Lifshitz transition at the LaAlO₃/SrTiO₃ interface”
S. Nandy, Narayan Mohanata, Swagata Acharya, and Arghya Taraphder, Phys. Rev. B **94**, 155103 (2016). (Citation: 13)

ARXIV/UNDER REVIEW

1. “Intrinsic Linear Response from Zeeman Quantum Geometry in 2D Unconventional Magnets”
Neelanjana Chakraborti, Sudeep Kumar Ghosh, **S. Nandy**, arXiv:2508.14745 (2025).
2. “Tunable Intraband and Interband Polarizability in three-dimensional nodal line semimetals”
Vivek Pandey, **S. Nandy**, Pankaj Bhalla, arXiv:2507.05232 (2023). (Under Review in NJP)
3. “Approximate half-integer quantization in anomalous planar transport in d-wave altermagnets”
Srimayi Korrapati, **S. Nandy**, and Sumanta Tewari, arXiv:2506.24122 (2025). (Accepted in PRB)
4. “Dark Matter Detection with Strongly Correlated Topological Materials: Flatband Effect”
Zhao Huang, Christopher Lane, Sarah E. Greife, **S. Nandy**, Benedikt Fauseweh, Silke Paschen, Qimiao Si, and Jian-Xin Zhu, arXiv:2305.19967 (2023). (Under Review in PRL)
5. “Electric field induced Berry curvature dipole and non-linear anomalous Hall effect in higher wave symmetric unconventional magnets”
Srimayi Korrapati, **S. Nandy**, and Sumanta Tewari, arXiv:2510.20237 (2025).

PROCEEDINGS

1. “*Revisiting quantum transport across junctions of single and double-Weyl semimetals*”
Suvendu Ghosh, **S. Nandy**, and Arghya Taraphder, Journal of Physics: Conference Series **2518**, 012005 (2023). (Citation: 1)
2. “*Effect of electron correlations on the Lifshitz transition at the $\text{LaAlO}_3/\text{SrTiO}_3$ interface*”
S. Nandy, Narayan Mohanata, Swagata Acharya, and Arghya Taraphder, AIP Conference Proceedings **1832**, 110026 (2017). (Citation: 1)

TALKS

- INVITED **Title:** Intriguing Transport Driven by Generalized Quantum Geometry in Unconventional Magnets,
September 2025, IIT Tirupati, India.
- Title:** Routes To Detect Band Topology,
October 2023, RRI, India.
- Title:** Weyl Semimetal and Its Intriguing Transport,
January 2022, CNLS, Los Alamos National Laboratory, USA.
- Title:** Intriguing Transport Properties in Weyl Semimetal,
December 2021, Delft University of Technology, Netherlands.
- Title:** Anomalous Transport Properties in Topological Systems,
December 2018, Weizmann Institute of Science, Israel.
- CONTRIBUTED **Title:** Quantum Hall Effect in a Weyl-Hubbard Model: Interplay between Topology and Correlations,
July 2023, SCES Meeting, South Korea.
- Title:** Weyl-Hubbard System: A Platform to Investigate the Interplay between Topology and Correlations,
March 2023, APS March Meeting, USA.
- Title:** Three Dimensional Quantum Hall Effect in Weyl-Hubbard Systems,
March 2023, CNLS Seminar, Los Alamos National Laboratory, USA.
- Title:** Chiral Anomaly Induced Non-linear Hall Effect in Multi-Weyl Semimetals,
July 2022, CNLS Seminar, Los Alamos National Laboratory, USA.
- Title:** Low-energy effective theory and anomalous Hall effect in monolayer WTe_2 ,
March 2022, APS March Meeting, USA.
- Title:** Theoretical Investigations on Anomalous Transport in Topological Systems,
January 2019, IIT Kharagpur, India.

POSTER PRESENTATIONS (CONFERENCES/SCHOOLS/WORKSHOPS)

- Title:** Quantum Hall effect in a Weyl-Hubbard model: interplay between topology and correlation,
November 2023, Conference: Topology, Symmetry and Interactions in Crystals: Emerging Concepts and Unifying Themes, KITP, Santa Barbara, USA.
- Title:** Planar Hall Effect in Weyl Semimetals,
August 2018, National Conference of Quantum Condensed Matter, Mohali, India.
- Title:** Anomalous transport near the Lifshitz transition at the $\text{LaAlO}_3/\text{SrTiO}_3$ interface,
June 2018, Summer School: Band structure meets quantum field theory, Vienna, Austria.
- Title:** Anomalous transport near the Lifshitz transition at the $\text{LaAlO}_3/\text{SrTiO}_3$ interface,
November 2017, Conference: Frontiers in Two-dimensional Quantum Systems, ICTP, Trieste, Italy.
- Title:** Change of Sign in the Magnetoresistance at $\text{LaAlO}_3/\text{SrTiO}_3$ Interface,
November 2016, International Conference on New Scintillations on Materials Horizon, Bareilly, India.
- Title:** Effect of electron correlations on the Lifshitz transition at the $\text{LaAlO}_3/\text{SrTiO}_3$ interface,
December 2016, DAE Solid State Physics Symposium, Odisha, India.

Winter School: January 2018, Topology and Interactions in Quantum Matter, IISER Kolkata, India.

Workshop: May 2017, Correlation and disorder in classical and quantum systems, IISC Bangalore, India.

Workshop: February 2016, Frontiers in Condensed Matter Physics, IOP Bhubaneswar, Odisha, India.

PROFESSIONAL SERVICES

2025-Present Technical Committee Member of the IEEE Magnetics Society.

2025-Present Early Career Editorial Board Member of Condensed Matter Journal.

2025 Coordinator of Pariksha Pe Charcha event at NIT Silchar.

2019-Present Referee for Physical Review Letters; Advanced Materials; Advanced Functional Materials; SciPost; Physical Review B (regular as well as Letter); Scientific Reports; Journal of Physics: Condensed matter; Europhysics Letters; European Journal of Physics B.

2017 Research Scholar Representative at the Department of Physics in IIT Kharagpur.

* Through online mode